AnsyTruss-2D User Manual

MATLAB-based 2D Truss Analysis Tool

Developed by: Dr. Vaishakh Kottila Veedu

Department of Mechanical Engineering

Auckland University of Technology (AUT)

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1 Introduction

AnsyTruss-2D is a **MATLAB-based** graphical user interface (GUI) app developed for the static analysis of two-dimensional pin-jointed trusses. The app provides a visual verification stage for checking the defined truss geometry, support conditions, and applied loads before carrying out the analysis using the direct stiffness method. Through editable input tables, users can define the truss geometry, material and sectional properties, support conditions, and applied nodal loads. The app then calculates nodal displacements, member axial forces, and support reactions, and provides visual outputs showing both the verified geometry and the deformed truss shape.

2 Coordinate System and Sign Convention

AnsyTruss-2D uses a global Cartesian coordinate system. The positive X-direction is defined from left to right, and the positive Y-direction is defined from bottom to top. Accordingly, nodal coordinates, applied loads, displacements, and support reactions should be interpreted using this sign convention.

3 Theory of 2D Truss Analysis

AnsyTruss-2D utilises the **Direct Stiffness Method** (Finite Element Method) to solve the truss system. For a 2D pin-jointed truss element connecting node i to node j with length L, cross-sectional area A, and Young's modulus E, the element stiffness matrix in the *local* coordinate system is given by:

$$\mathbf{k} = \frac{EA}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{1}$$

To relate the local coordinates to the global coordinate system, a transformation matrix $\mathbf{T}$ is applied:

$$\mathbf{T} = \begin{bmatrix} c & s & 0 & 0 \\ -s & c & 0 & 0 \\ 0 & 0 & c & s \\ 0 & 0 & -s & c \end{bmatrix} \tag{2}$$

where $c = \cos\theta = \frac{x_j - x_i}{L}$, $s = \sin\theta = \frac{y_j - y_i}{L}$, and the element length is $L = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$.

The element stiffness matrix in the *global* coordinate system ($\mathbf{K}^e$) is obtained by pre- and post-multiplying the local stiffness matrix by the transformation matrix ($\mathbf{K}^e = \mathbf{T}^T\mathbf{k}\mathbf{T}$), which expands to:

$$\mathbf{K}^e = \frac{EA}{L}\begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix} \tag{3}$$

The global stiffness matrix for the entire truss ($\mathbf{K}$) is assembled by summing the contributions of all N_e individual elements:

$$\mathbf{K} = \sum_{e=1}^{N_e} \mathbf{K}^e \tag{4}$$

Once boundary conditions (supports) are applied to remove rigid body motion, the unknown nodal displacements ($\mathbf{U}$) are solved using the global equilibrium equation:

$$\mathbf{K}\mathbf{U} = \mathbf{F} \tag{5}$$

where $\mathbf{F}$ is the global vector of applied nodal forces.

Finally, the internal axial force f developed within each member is extracted from the computed global displacements at its start node i and end node j:

$$f = \frac{EA}{L} \begin{bmatrix} -c & -s & c & s \end{bmatrix} \begin{Bmatrix} U_{ix} \\ U_{iy} \\ U_{jx} \\ U_{jy} \end{Bmatrix} \tag{6}$$

A positive calculated value for f indicates the member is in tension, while a negative value indicates compression.

4 App Panels

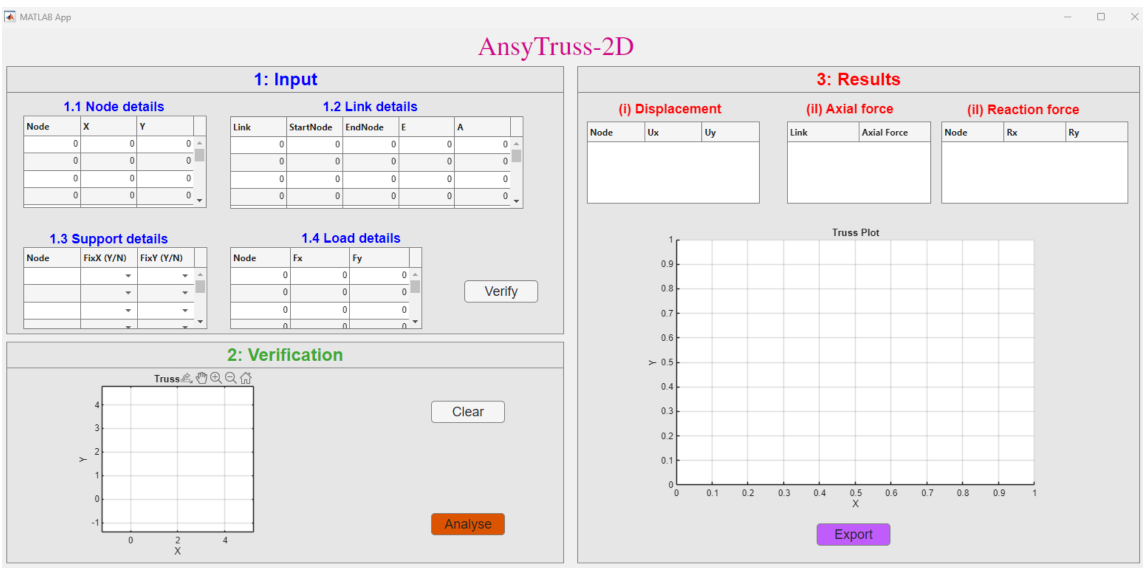

Figure 1: Overview of the AnsyTruss-2D graphical user interface showing the main app panels.

The GUI is organised into three main panels: the **Input Panel**, the **Output Panel**, and the **Visualisation/Verification Panel**. These panels are arranged to provide a clear workflow for defining the truss model, verifying the entered geometry and boundary conditions, and reviewing the analysis results.

4.1 Input Panel

The Input Panel is used to define the truss model. It allows the user to enter the required model data, including node coordinates, link connectivity, support conditions, and applied loads. The four main components are:

- Node Data
- Link Data
- Support Data
- Load Data

Users can type the required values directly into the tables. Blank rows may be left unused.

4.2 Verification Panel

The Verification Panel is used to check the truss model before analysis. After clicking the **Verify** button, the app plots the initial truss geometry and displays the entered model visually. The verification plot displays:

- Node numbers
- Link numbers
- Pinned and roller support symbols
- Applied loads with appropriate directional arrows

This step helps confirm that the geometry, connectivity, support conditions, and applied loads have been entered correctly before running the analysis. The **Clear** button is used to clear the entered input data from the input tables.

4.3 Results Panel

The Results Panel displays the outputs of the truss analysis. After clicking the **Analyse** button, the app shows:

- Nodal displacement table
- Member axial force table
- Reaction force table
- Deformed truss shape overlaid on the initial undeformed geometry, with axial force contours displayed along the members

The **Export** button is provided to generate and save the analysis report in PDF format.

5 Preloaded Example and Input Data Format

A simple three-node triangular truss is provided as a preloaded demonstration example in the software (Figure 2). This example is useful for quickly understanding the data format and testing the **Verify**, **Analyse**, and **Export PDF Report** functions.

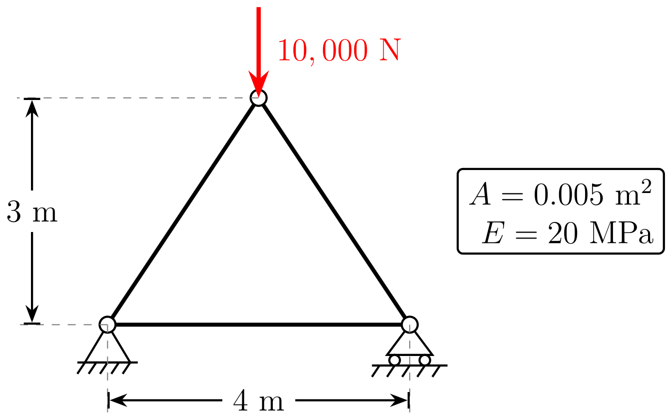

Figure 2: A 2D Pin-Jointed triangular truss with cross-sectional area $A = 0.005$ m^2 and Young's modulus $E = 20$ MPa.

5.1 Fixing the Coordinates

To mathematically model the truss, we must establish a global coordinate system. Let us place the origin $(0, 0)$ at the bottom-left pin support. Because the truss has a 4 m span and is symmetric, the top node is located at the horizontal midpoint ($X = 2$ m) and a height of $Y = 3$ m.

5.2 Assigning Node and Link Numbers

Before entering data, you must assign a unique **Node number** to every joint and a unique **Link number** to every member. For this example, we assign:

- **Node 1:** Bottom-left support
- **Node 2:** Bottom-right support
- **Node 3:** Top apex

These assignments dictate how the following input tables are filled out.

5.3 Node Data Format

The Node Data table defines the position of each node based on your coordinate system.

Node	X	Y
1	0.0	0.0
2	4.0	0.0
3	2.0	3.0

Columns: `Node` (Node number), `X` (Horizontal coordinate), `Y` (Vertical coordinate).

5.4 Link Data Format

The Link Data table defines the truss members and their connectivity.

Link	StartNode	EndNode	E	A
1	1	2	2.0×10^7	0.005
2	1	3	2.0×10^7	0.005
3	2	3	2.0×10^7	0.005

Columns: `Link` (Element number), `StartNode` and `EndNode` (The nodes it connects), `E` (Young's modulus), `A` (Cross-sectional area).

5.5 Support Data Format

The Support Data table defines restrained and free degrees of freedom.

Node	FixX	FixY
1	Y	Y
2	N	Y

Use `Y` to signify a restrained (fixed) degree of freedom, and `N` for a free degree of freedom. For example, Node 1 is a pin (`Y, Y`) and Node 2 is a horizontal roller (`N, Y`).

5.6 Load Data Format

The Load Data table defines nodal loads.

Node	Fx	Fy
3	0.0	-10000.0

Sign convention: Positive `Fx` acts to the right, negative `Fx` acts to the left. Positive `Fy` acts upward, negative `Fy` acts downward.

6 Using the App

Step 1: Enter Input Data

Enter your mapped node data, link data, support data, and load data into the corresponding tables in the Input Panel (see Figure 3). **Use consistent units throughout the model**. For example, if length is entered in metres and force in Newtons, then Young's modulus and area must also be entered in compatible units (Pascals and m^2).

Step 2: Verify the Truss

Click the **Verify** button. The app checks the input data and plots the verification geometry (see Figure 4). The verification stage checks for empty tables, duplicate nodes, invalid start/end nodes, zero-length links, and invalid support/load locations. If the displayed geometry does not appear correct, return to the input tables and edit the data.

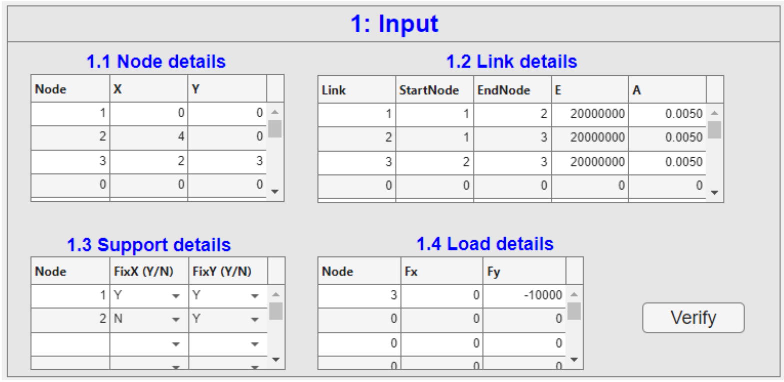

Figure 3: Input panel populated with the data from the preloaded example.

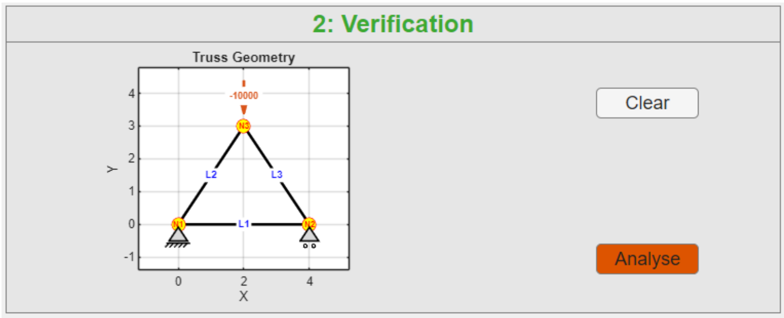

Figure 4: Verification panel showing Node numbers, Link numbers, supports, and applied loads.

Step 3: Analyse the Truss

Click the **Analyse** button. The app solves the truss and displays the results in the Results Panel (see Figure 5). The final plot shows the deformed truss shape with an axial force contour. Positive axial force indicates tension, while negative axial force indicates compression.

Step 4: Export PDF Report

Click the **Export PDF Report** button to generate an A4 PDF report. Wait for the pop-up message saying **"PDF report exported successfully."** before opening the file.

7 Output Results

7.1 Displacement

The nodal displacement table contains **Ux** (horizontal displacement) and **Uy** (vertical displacement).

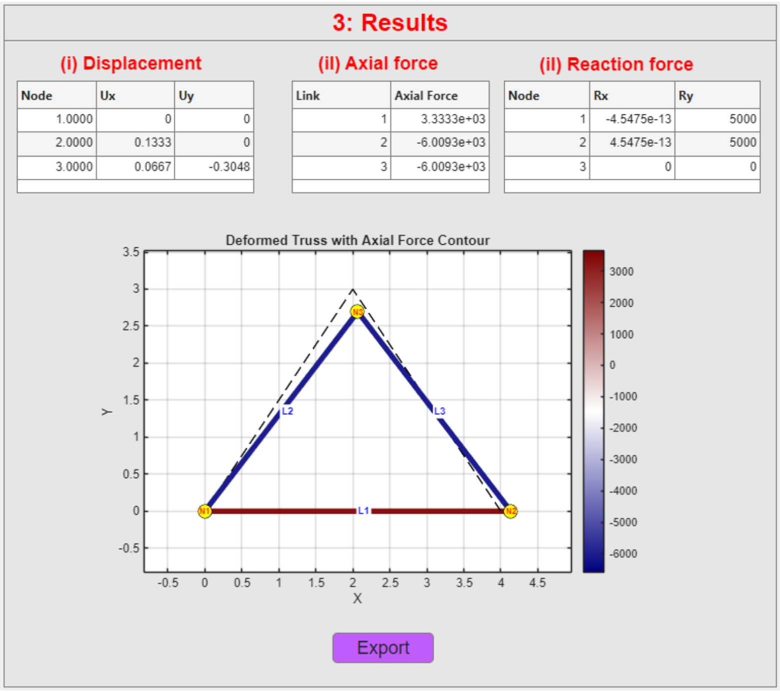

Figure 5: Results panel displaying tabular outputs alongside the deformed and undeformed configurations.

7.2 Axial Force

The axial force table outputs the internal force per link. Positive force indicates tension, and negative force indicates compression.

7.3 Reaction Force

The reaction table contains **Rx** (horizontal reaction) and **Ry** (vertical reaction).

8 Assumptions and Limitations

AnsyTruss-2D is developed for the static analysis of two-dimensional, pin-jointed truss structures. The following assumptions are adopted:

- The structure is a 2D pin-jointed truss and is in static equilibrium.
- Members carry axial force only; bending moments, shear forces, and torsional effects are not considered.
- Material behaviour is linear elastic and deformations are assumed to be small.
- Loads are applied only at nodes and connections are ideal frictionless pins.
- Geometric nonlinearity, material nonlinearity, buckling, and dynamic effects are not considered.
- Units must be consistent throughout the input data.

9 Troubleshooting

Problem	Possible Solution
Verification fails	Check that node numbers are unique and that all link start/end nodes exist.
Analysis does not run	Check supports, loads, and ensure the truss is stable.
No load arrow appears	Check that the load node exists and that Fx or Fy is not zero.
Unexpected displacement values	Check units for E, A, length, and force.
PDF does not open correctly	Wait for the export-success pop-up before opening the file.
Link table overflows in report	Use compact column values or reduce the number of visible rows.

10 Summary

AnsyTruss-2D provides a simple and structured workflow for defining, verifying, analysing, visualising, and reporting two-dimensional truss structures. As a MATLAB-based GUI tool, it supports static analysis of linear elastic pin-jointed trusses using the direct stiffness method. It is developed at Auckland University of Technology (AUT) as an educational and engineering calculation tool for truss analysis.